

8

ELECTROMAGNETISM AND EM WAVES

CHAPTER OUTLINE

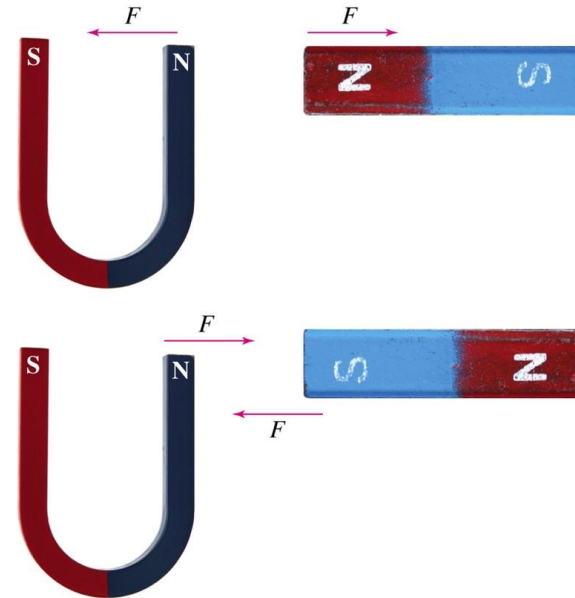
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Donald J. Bord



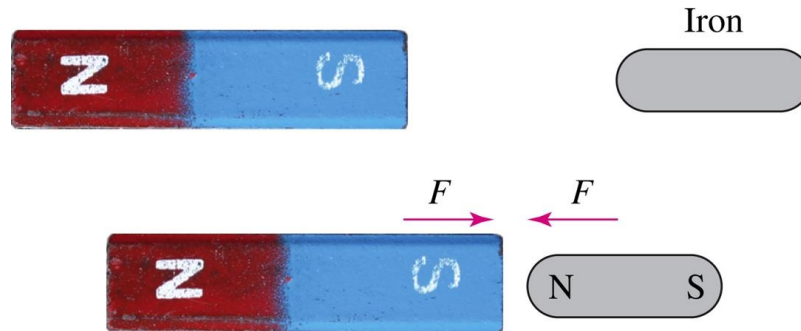
8.1 Magnetic Materials and Fields

- ❑ All simple magnets exhibit one end is attracted to the north, and the opposite end is attracted to the south.
- ❑ The north-seeking part of a magnet is called its north pole, and the south-seeking part is its south pole.
- ❑ All magnets have both poles. If a magnet is broken into pieces, each part will have its own north and south poles.
- ❑ The south pole of one magnet exerts a mutually attractive force on the north pole of a second magnet. The south poles of two magnets repel.



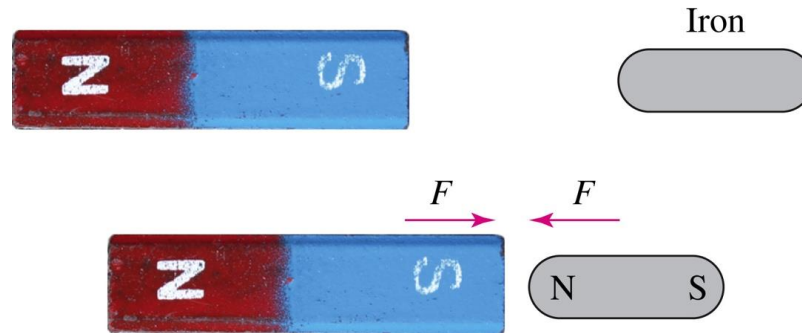
Like poles repel; unlike poles attract

8.1 Magnetic Materials and Fields



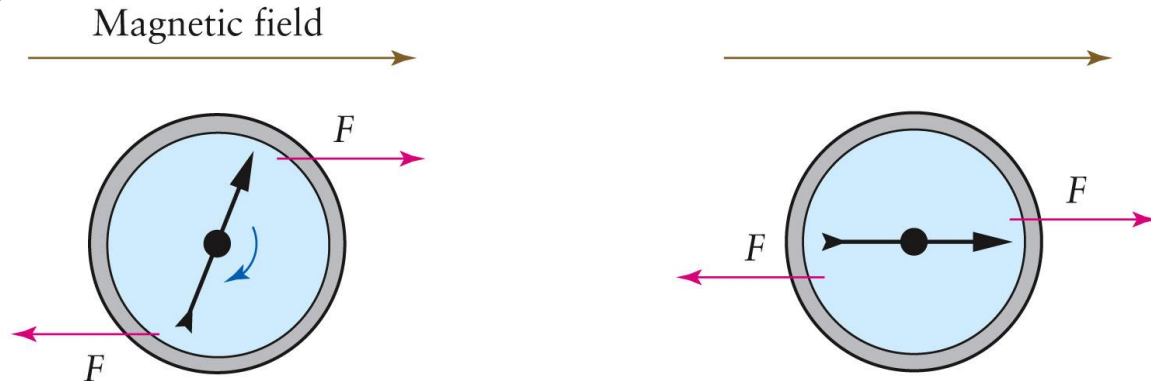
- ❑ Metals that are strongly attracted by magnets are said to be *ferromagnetic*.
- Such materials have magnetism induced in them when they are near a magnet.

8.1 Magnetic Materials and Fields



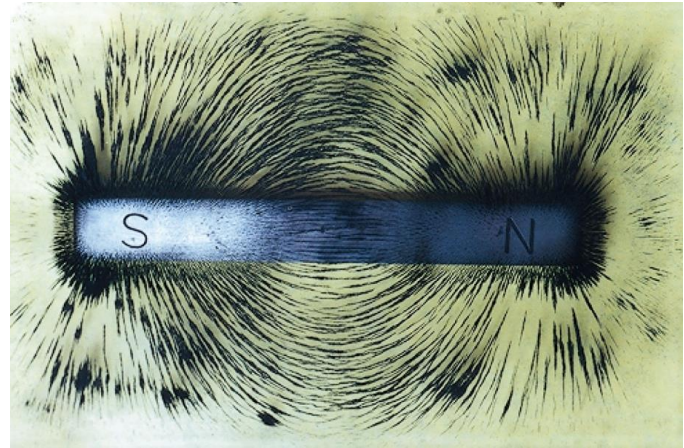
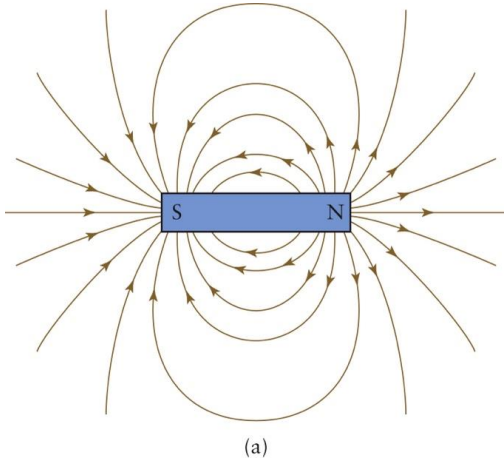
- If a piece of iron is brought near the south pole of a magnet, the part of the iron nearest the magnet has a north pole induced in it, and the part farthest away has a south pole induced in it.
- Once the iron is removed from the vicinity of the magnet, it loses most of the induced magnetism.
- Some ferromagnetic metals actually retain the magnetism induced in them—they become *permanent magnets*.

8.1 Magnetic Materials and Fields



- ❑ A magnetic field is produced by a magnet and acts as the agent of the magnetic force.
 - The poles of a second magnet experience forces when in the magnetic field: Its north pole has a force in the same direction as the magnetic field, but its south pole has a force in the opposite direction.
 - A compass can be thought of as a “magnetic field detector” because its needle will always try to align itself with a magnetic field.

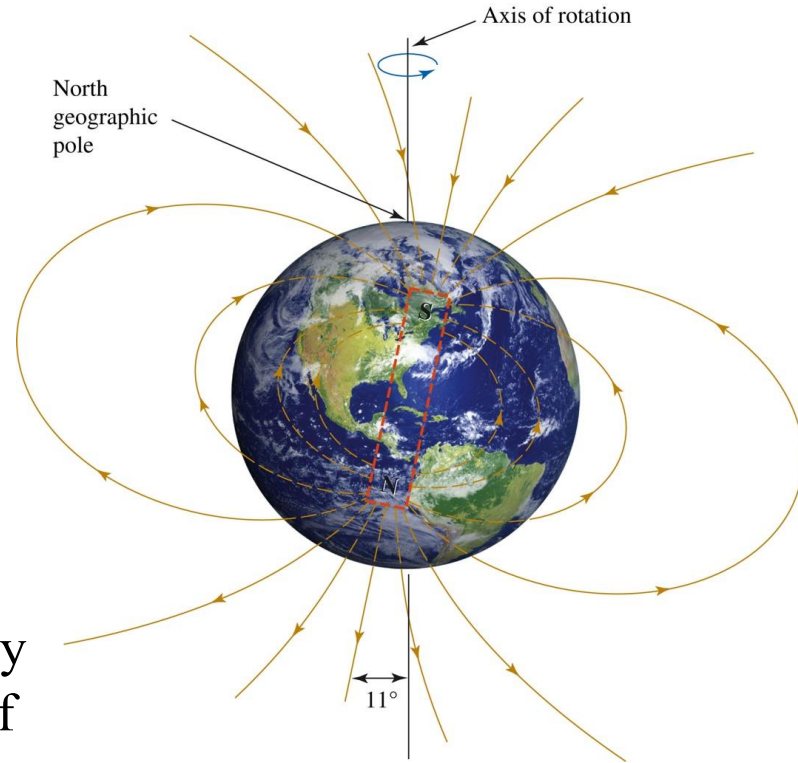
8.1 Magnetic Materials and Fields



- ❑ The direction of a field line at a particular place is the direction in which the north pole of a compass needle at that location points.
- ❑ *The magnetic field lines point toward the south pole and away from the north pole.*

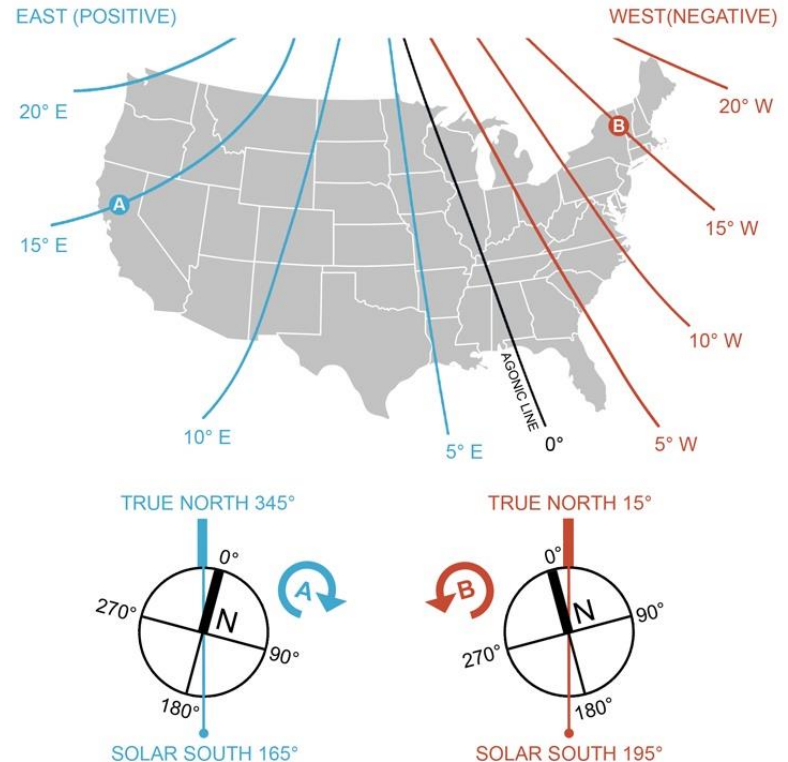
8.1 Earth's Magnetic Field

- ❑ Because magnets respond to magnetic fields, the fact that compass needles point north indicates that Earth itself has a magnetic field.
- ❑ Earth's magnetic field has the same general shape as the field around a bar magnet, with its poles tilted about 11° with respect to the axis of rotation.
- ❑ The direction of “true north” shown on maps is determined by the orientation of Earth's axis of rotation. The axis is aligned closely with Polaris, the North Star



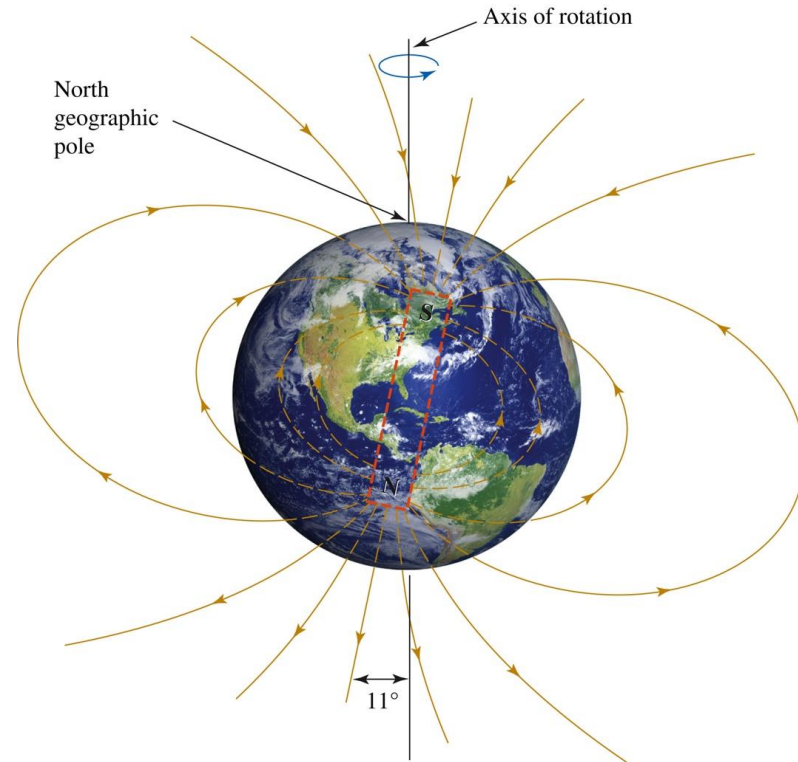
8.1 Earth's Magnetic Field

- Because of the tilt of Earth's "magnetic axis," at most places on Earth compasses do not point to true north. For example, in the western two-thirds of the United States, compasses point to the right (east) of true north, whereas in New England compasses point to the left (west) of true north.



8.1 Earth's Magnetic Field

- ❑ The difference, in degrees, between the direction of a compass and the direction of true north varies from place to place and is referred to as the *magnetic declination*.



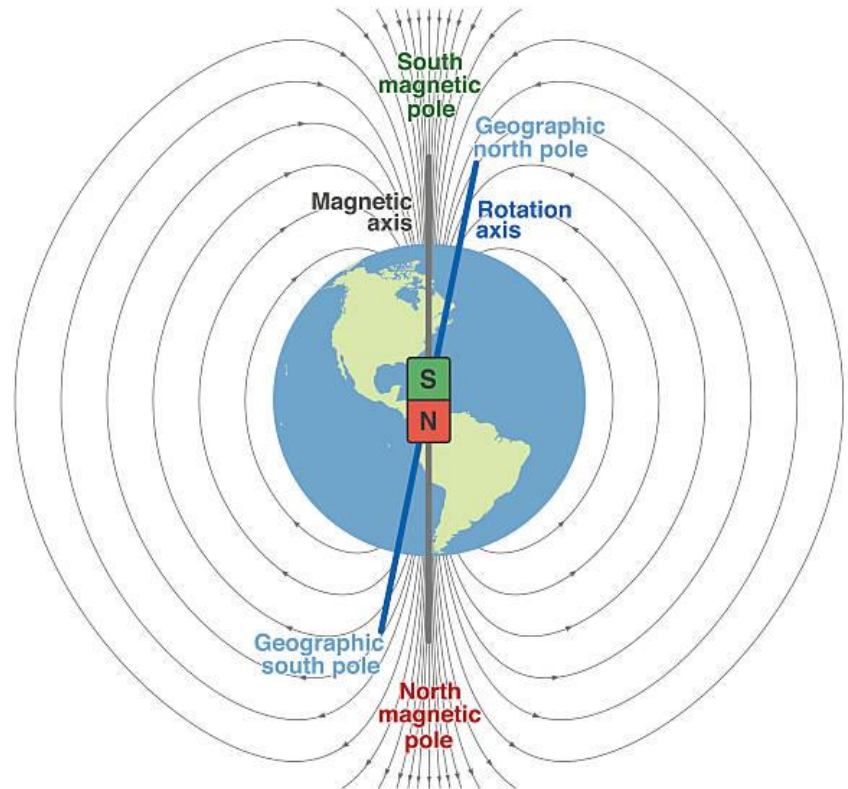
8.1 Earth's Magnetic Field

Another thing to note about Earth's magnetic field: Earth's north magnetic pole is at (near) its south geographic pole, and vice versa. Why?

1. The north pole of a magnet is attracted to the south pole of a second magnet.

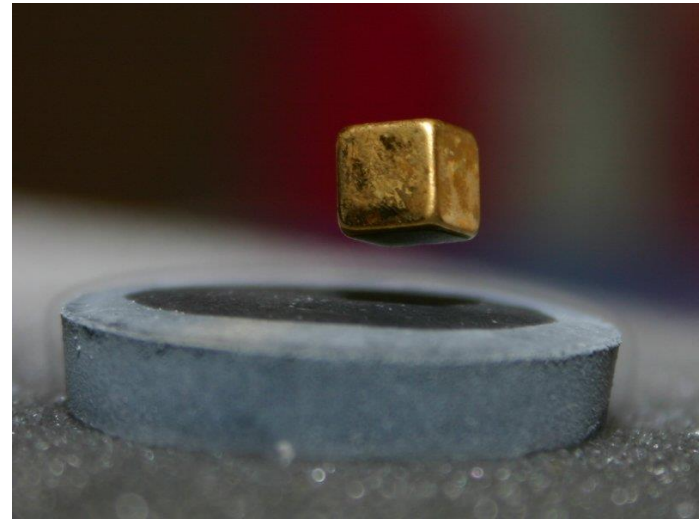
2. The north pole of a compass needle points to the north.

Therefore, a compass's north pole points at the Earth's south magnetic pole.



8.1 Earth's Magnetic Field

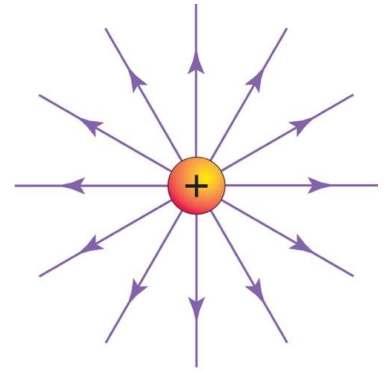
- ❑ Superconductors, so named because of their ability to carry electric current with zero resistance, react to magnetic fields in a rather startling fashion.
- ❑ In the superconducting state, the material will expel any magnetic field from its interior. This phenomenon, known as *the Meissner effect*, is why strong magnets are levitated when placed over a superconductor.
- ❑ When trying to determine whether a material is in the superconducting state, it is easier to test for the presence of the Meissner effect than it is to see if the resistance is exactly zero.



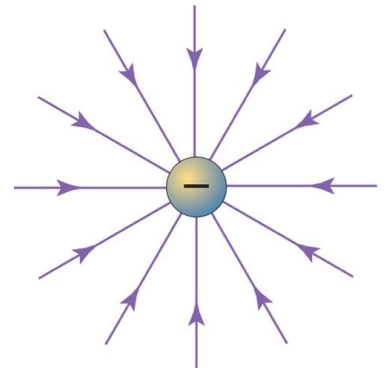
8.2 Interactions between Electricity and Magnetism

Let's summarize and review the key aspects of electrostatics and magnetism:

- ❑ Electric charges produce electric fields in the space around them.
- ❑ An electric field causes a force on any charged object placed in it.

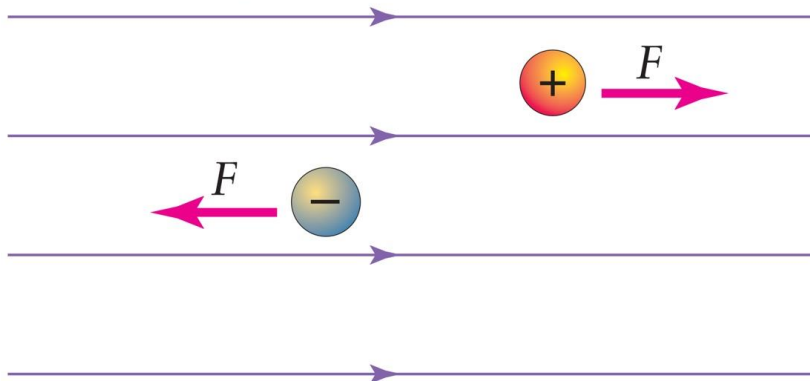


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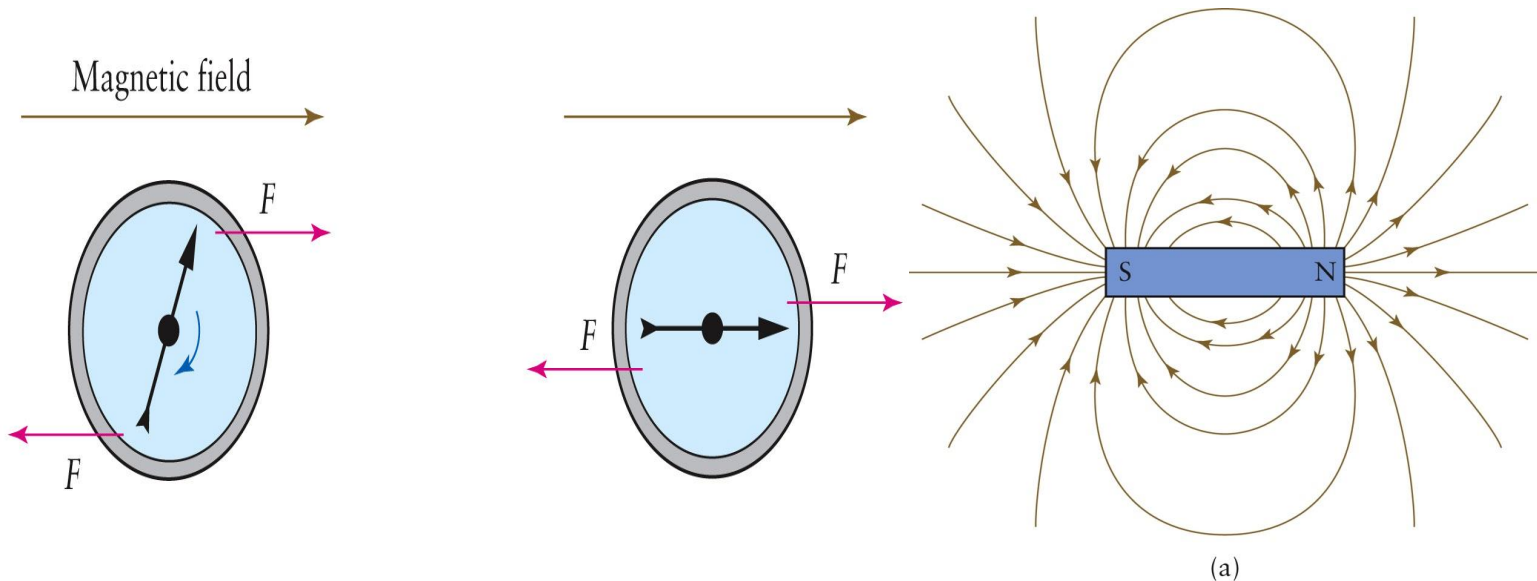
(b)

Electric field



8.2 Interactions between Electricity and Magnetism

- ❑ Magnets produce magnetic fields in the space around them.
- ❑ A magnetic field causes forces on the poles of any magnet placed in it.



8.2 Electromagnets: Observation 1

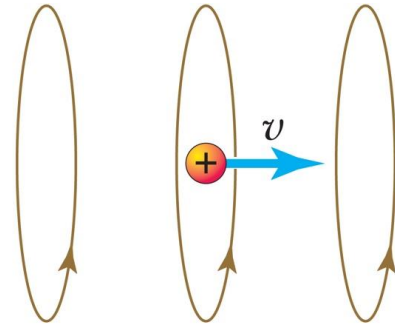
We can make *three observations regarding electromagnetism*.

Observation 1: A moving electric charge produces a magnetic field in the space around it. An electric current produces a magnetic field around it.

- ❑ The first observation is the basis of electromagnets.

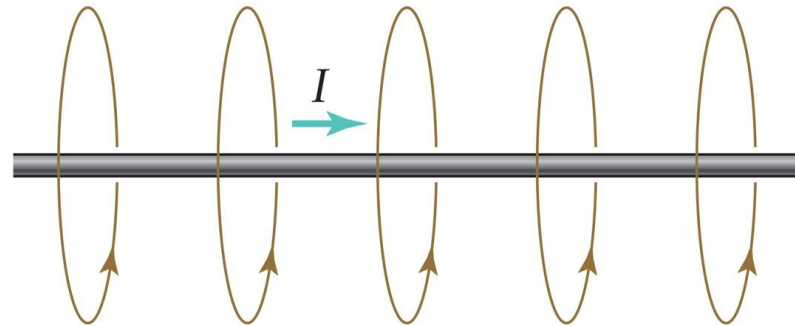
8.2 Electromagnets: Observation 1

- ❑ A single charged particle creates a magnetic field only when it is moving.



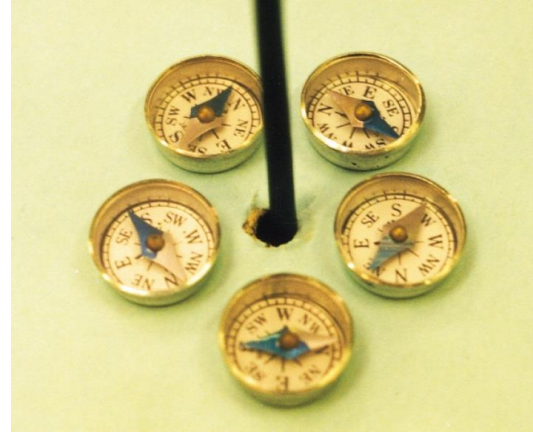
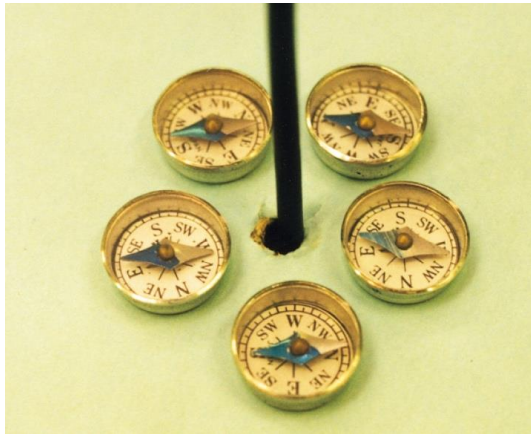
(a)

- ❑ The magnetic field produced is in the shape of circles around the path of the charge.



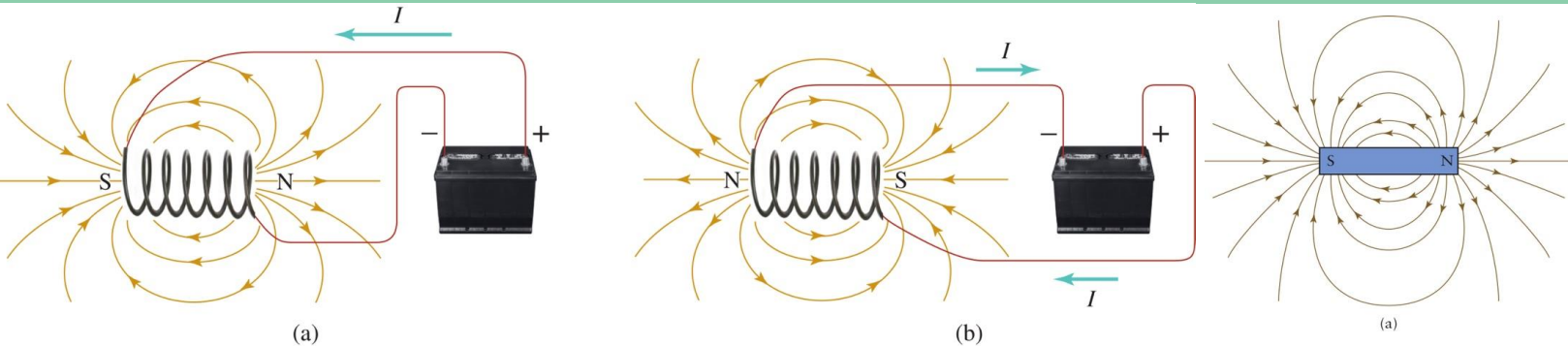
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8.2 Electromagnets: Observation 1



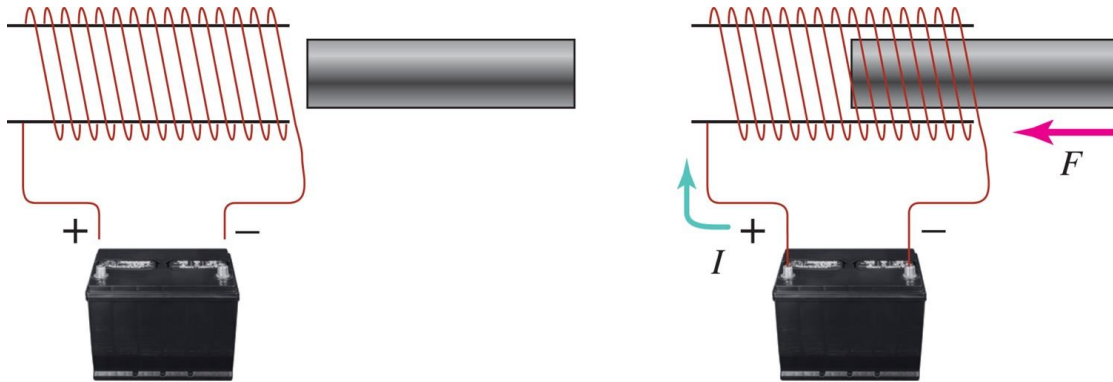
- For a steady (DC) current, which is basically a succession of moving charges, in a wire, the field is steady, and its strength is proportional to the size of the current and inversely proportional to the distance from the wire.
- Reversing the direction of the current in the wire will reverse the directions of the magnetic field lines.

8.2 Electromagnets: Observation 1



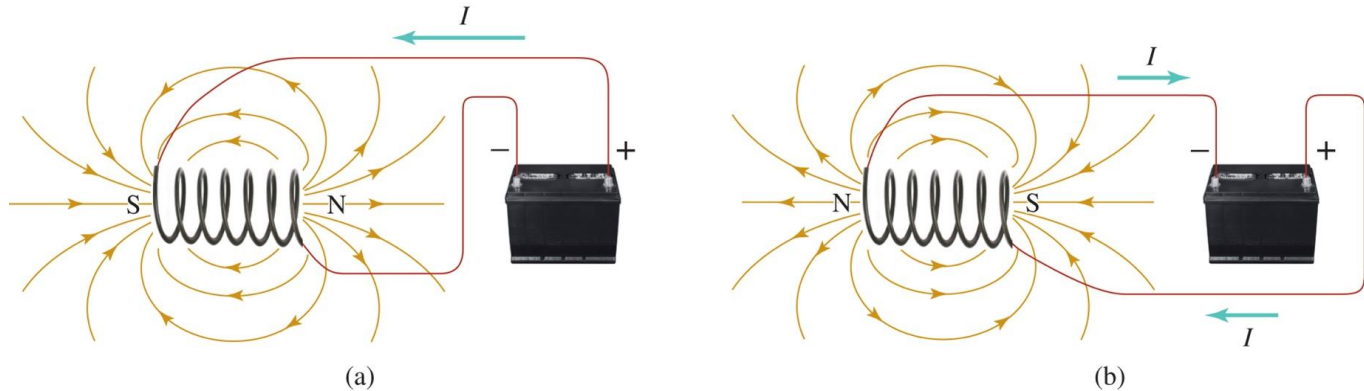
- Most applications of this phenomenon use *coils*—long wires wrapped in the shape of a cylinder, often around an iron core.
- The magnetism induced in the iron greatly enhances the magnetic field of the coil.
- The magnetic field of such a coil (when carrying a direct current) has the same shape as the field around a bar magnet. This device is an electromagnet.
- It behaves just like a permanent magnet as long as there is a current flowing. One end of the coil is a north pole, and the other is a south pole.

8.2 Electromagnets: Observation 1



- A coil with a length that is much greater than its diameter is called a *solenoid*.
- If an iron rod is partially inserted into a solenoid with a hollow core, the rod will be pulled in when the current is switched on: The magnetic pole associated with the coil's field nearest the rod induces a magnetic field with the opposite polarity in the rod, thus exerting an attractive force on it.
- Solenoids are used in common devices such as striking doorbell chimes and engaging starter motors on car and truck engines.

8.2 Electromagnets: Observation 1



- ❑ The polarity of an electromagnet is reversed if the direction of the current is reversed.
- ❑ An alternating current (AC) in a coil will produce a magnetic field that oscillates: It increases, decreases, and switches polarity with the same frequency as the current.
- ❑ Such an oscillating magnetic field will cause a nearby piece of iron to vibrate.

8.2 Electromagnets: Observation 1

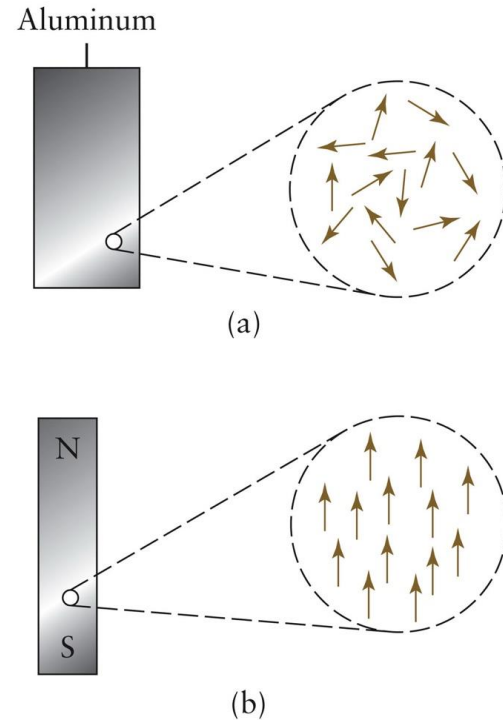
- ❑ Electromagnets are used to produce the strongest magnetic fields on Earth.
- ❑ Two factors contribute to stronger fields: 1) wrapping more coils around the cylinder and 2) using a larger electric current.
 - Wrapping more coils suggests the use of thinner wire so that more coils can fit into the same amount of space.
 - But smaller wire requires smaller electric current so that the wire does not overheat and melt.

8.2 Electromagnets: Observation 1

- This limitation of smaller current is overcome in *superconducting electromagnets*.
- When the wire used in an electromagnet is a superconductor, it can carry huge electric currents with no ohmic heating because there is no resistance.
- Very small superconducting electromagnets can generate very strong magnetic fields while using much less electrical energy than a conventional electromagnet.
- Superconducting electromagnets do have limitations, though. The superconducting state is lost if the temperature, electric current, or magnetic field strength exceeds certain values.

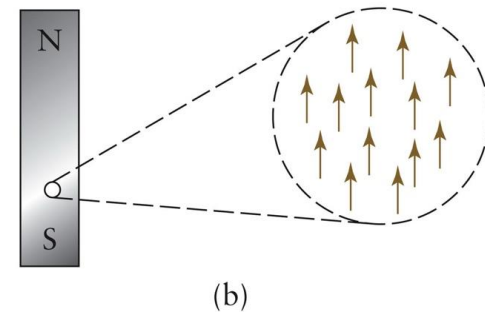
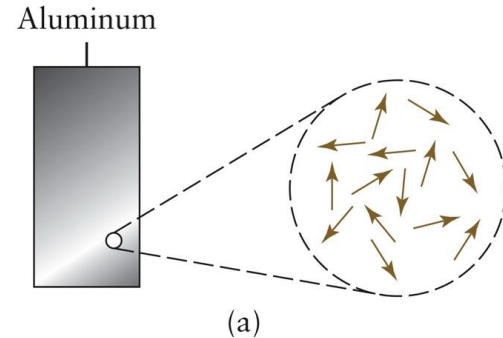
8.2 Electromagnets: Observation 1

- ❑ This first interaction not only explains how electromagnets work, but also gives us new insight into permanent magnets.
- Because electrons in atoms are charged particles in motion about the nucleus, they produce magnetic fields.
- The electrons have their own magnetic fields associated with their spin.



8.2 Electromagnets: Observation 1

- In any unmagnetized material, the individual magnetic fields of the electrons are randomly oriented and cancel each other out (a).
- In ferromagnetic materials, these fields can be aligned with one another by an external magnetic field (b). The material then produces a net magnetic field.



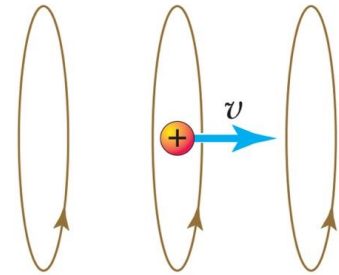
8.2 Electromagnets: Observation 2

Observation 2: A magnetic field exerts a force on a moving electric charge. Therefore, a magnetic field exerts a force on a current-carrying wire.

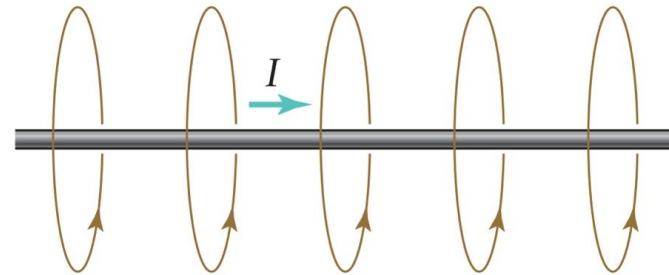
- The second observation helps us understand how things such as electric motors and speakers work.

8.2 Electromagnets: Observation 2

- ❑ A stationary electric charge is not affected by a magnetic field, but a moving charge usually is.
- ❑ The direction of the magnetic field from a current-carrying wire is perpendicular to the direction the current is flowing.
- ❑ If a charge's velocity or the direction of a current is parallel to that of the magnetic field or in the opposite direction, the magnetic field does not exert this force.

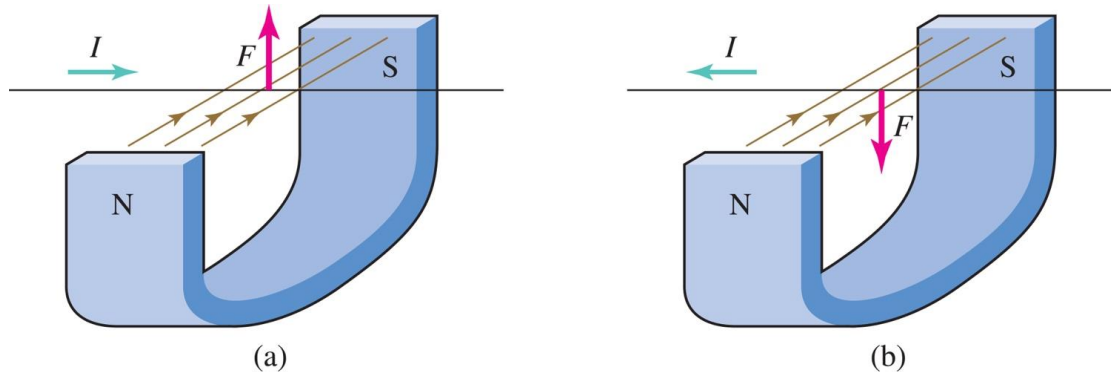


(a)



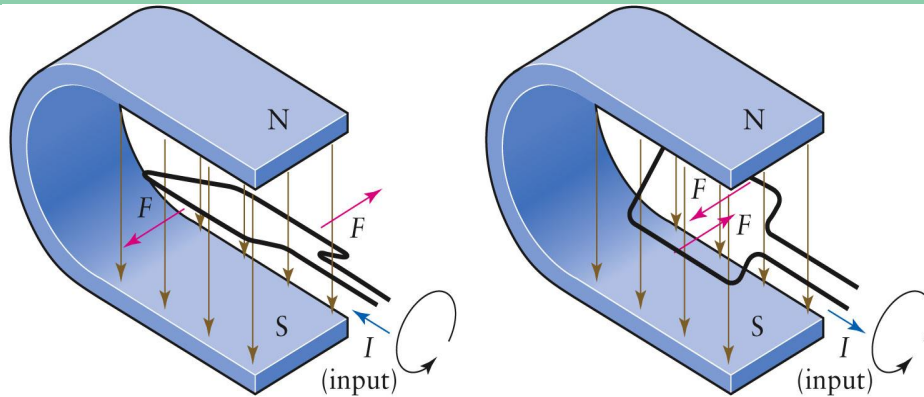
(b)

8.2 Electromagnets: Observation 2



- ❑ Similarly, the force that a magnetic field exerts on a moving charge or on a current-carrying wire is *perpendicular* to both the direction of the magnetic field and the direction the charge is flowing.
- ❑ For example, if a horizontal magnetic field is directed away from you and a wire is carrying a current to your right, the force on the wire is upward.
- ❑ If the direction of the current is reversed, the direction of the force is reversed (downward).
- ❑ An alternating current would cause the wire to experience a force that alternates up and down.

8.2 Electromagnets: Observation 2



- *Electric motors*—like those in power drills and elevators—exploit observation 2. The simplest type of electric motor consists of a coil of wire mounted so that it can rotate in the magnetic field of a horseshoe-shaped magnet.
- A direct current flows through the coil, the magnetic field causes forces on the sides of the coil, and the coil rotates. Once the coil has completed half of a rotation, a simple mechanism reverses the direction of the current. This reverses the force on the coil, causing it to rotate another half turn.
- This process is repeated, and the coil spins continuously.

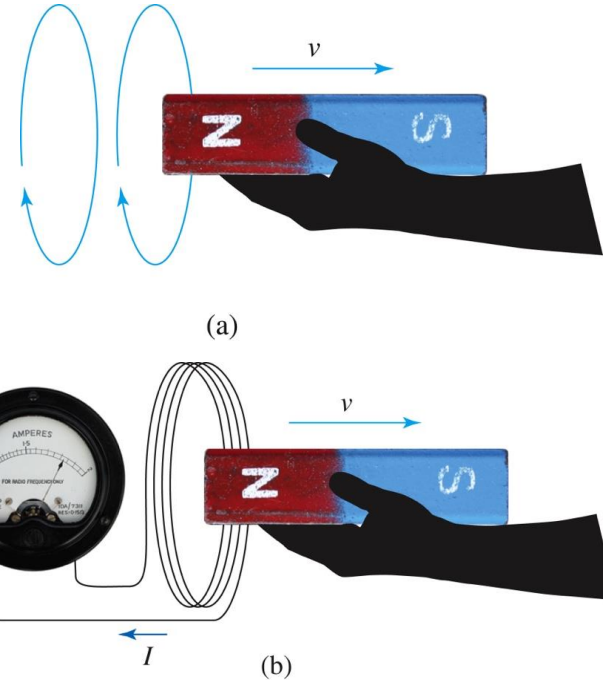
8.2 Electromagnetic Induction: Observation 3

Observation 3: A moving magnet produces an electric field in the space around it. A coil of wire moving through a magnetic field has a current induced in it.

- The third observed interaction between electricity and magnetism is called: *Electromagnetic Induction* and is used by electric generators.

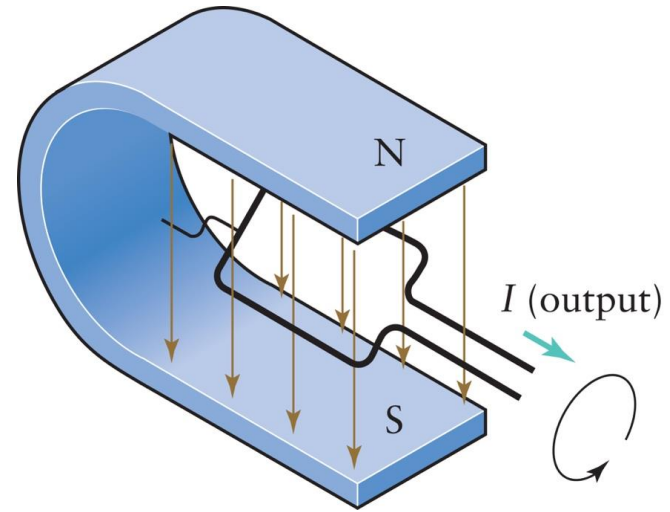
8.2 Electromagnetic Induction: Observation 3

- The electric field around a moving magnet is in the shape of circles around the path of the magnet.
- This circular electric field will force charges in a coil of wire to move in the same direction—as a current.
- The process of inducing an electric current with a magnetic field is known as *electromagnetic induction*.
- All that is required is that the magnet and coil move relative to each other. If the coil moves and the magnet remains stationary, a current is induced.
- If the motion is steady in either case, the induced current is in one direction. If either the coil or the magnet oscillates back and forth, the current alternates with the same frequency—it is AC.

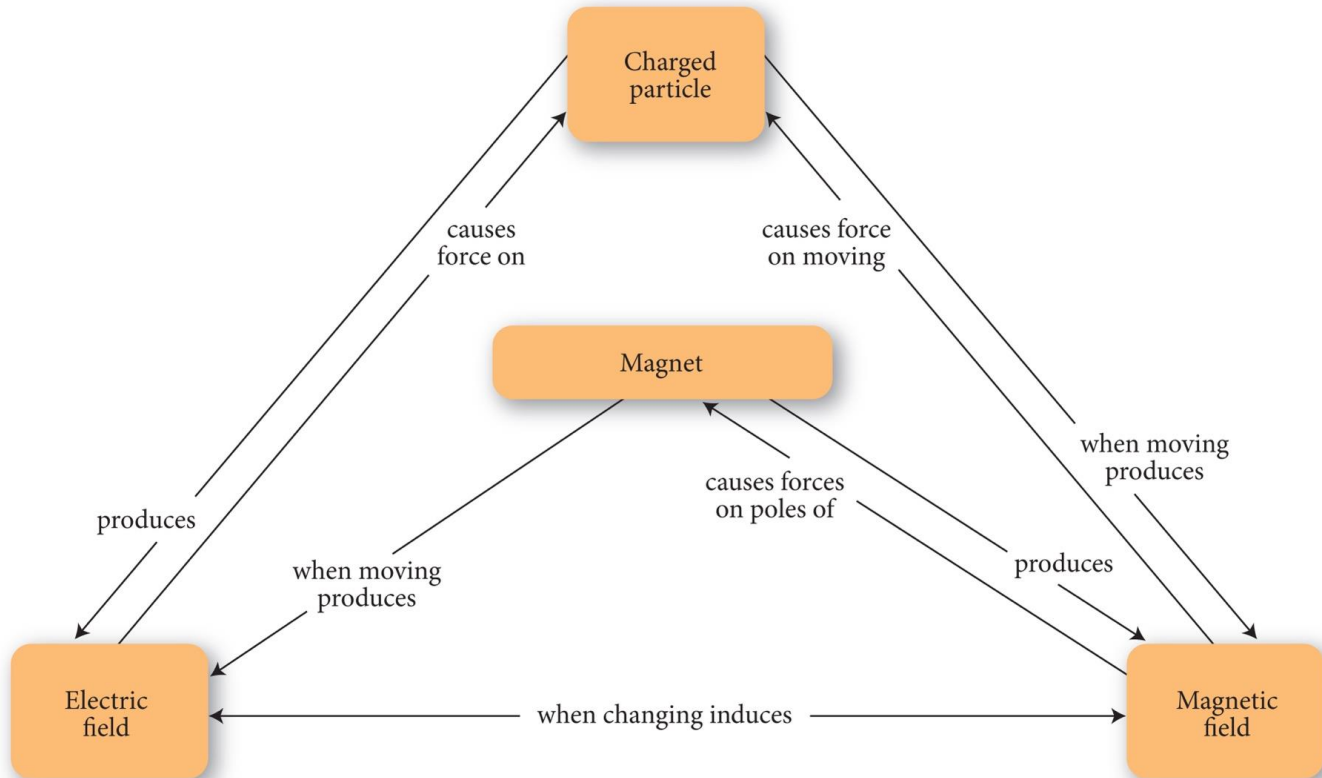


8.2 Electromagnetic Induction: Observation 3

- ❑ Electromagnetic induction is used in the most important device to produce electricity: the *generator*.
- ❑ When the coil is forced to rotate, it moves relative to the magnet, so a current is induced in it.
- ❑ We might call this device a “two-way energy converter.”:
 - When electrical energy is supplied to it, it is a motor. It converts this electrical energy into mechanical energy of rotation.
 - When mechanical energy is supplied to it by mechanically turning it, it is a generator. It converts this mechanical energy into electrical energy.



Concept Map



8.3 Principles of Electromagnetism

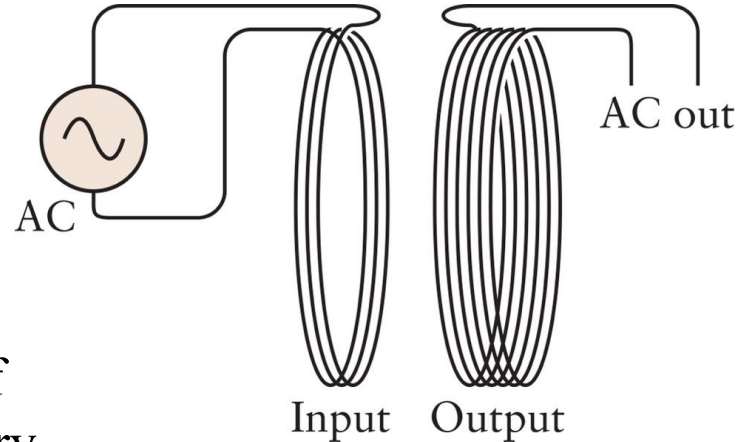
The interactions between electricity and magnetism we just discussed, suggest the following two general statements. We might call these the *principles of electromagnetism*:

1. An electric current or a changing electric field induces a magnetic field.
 2. A changing magnetic field induces an electric field.
- These two statements summarize the previous observations. In both cases, a “changing” field means that the strength or the direction (or both) of the field is changing.

8.3 Principles of Electromagnetism

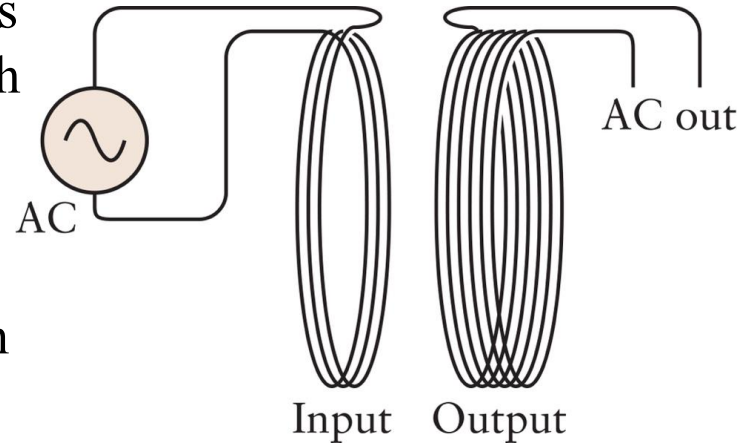
A **transformer** is a device used to step-up or step-down AC voltages.

- A transformer consists of two separate coils of wire in close proximity.
- An AC voltage is applied to one of the coils, called the input or primary coil, and an AC voltage appears at the other coil, called the output or secondary coil.



8.3 Principles of Electromagnetism

- The AC in the primary coil produces an oscillating magnetic field through both coils.
- This oscillating (and therefore changing) magnetic field induces an AC current in the output coil.



- ❑ Transformers do not work with DC. As a DC input would produce a steady magnetic field that would not induce a current in the output coil.

8.3 Principles of Electromagnetism

Now, in transformers, how can the voltage of the output be different from the voltage of the input?

Each “turn” of the output coil has the same voltage induced in it. The voltages in all of the turns add together so that the more turns there are in the output coil, the higher the total voltage. The ratio of the number of turns in the two coils determines the ratio of the input and output voltages.

$$\frac{\text{voltage of output}}{\text{voltage of input}} = \frac{\text{number of turns in output coil}}{\text{number of turns in input coil}}$$

$$\frac{V_o}{V_i} = \frac{N_o}{N_i}$$

Example 8.1

A transformer is being designed to have a 600-volt output with a 120-volt input. If there are to be 800 turns of wire in the input coil, how many turns must there be in the output coil?

$$\frac{V_o}{V_i} = \frac{N_o}{N_i}$$

$$\frac{600 \text{ V}}{120 \text{ V}} = \frac{N_o}{800}$$

$$800 \times 5 = N_o$$

$$N_o = 4,000 \text{ turns}$$

8.5 Electromagnetic Waves

Eyes, radios, televisions, radar, x-ray machines, microwave ovens, heat lamps. . . What do all of these things have in common?

They all use electromagnetic waves (EM waves).

EM waves occupy prominent places both in our daily lives and in our technology. These waves are also involved in many natural processes and are essential to life itself.

Now we will discuss the nature and properties of electromagnetic waves and look at some of their important roles in today's world.

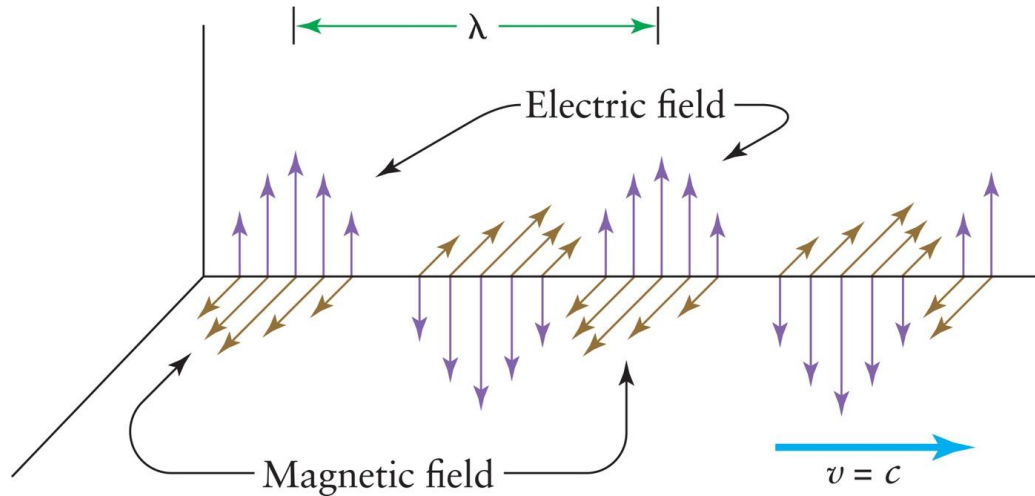
8.5 Electromagnetic Waves

EM waves involve both electricity and magnetism.

DEFINITION

Electromagnetic Wave: A transverse wave consisting of a combination of oscillating electric and magnetic fields.

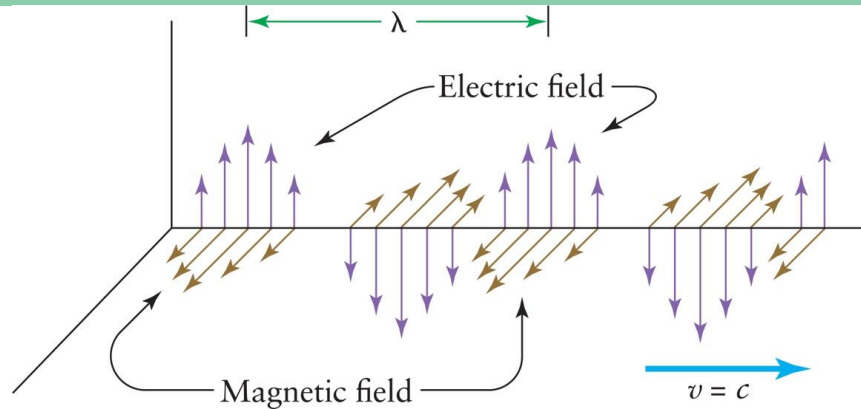
8.5 Electromagnetic Waves



The figure shows “snapshots” of an EM wave traveling to the right. The three axes are perpendicular to each other.

In this particular case, the electric field is vertical. As the wave travels by a given point in space, the electric field oscillates up and down, the magnetic field at the point oscillates horizontally in and out.

8.5 Electromagnetic Waves



- ❑ Electromagnetic waves do differ from mechanical or sound waves in two important ways:
- First, they are a combination of two waves in one: an electric field wave and a magnetic field wave. These cannot exist separately.
 - Second, EM waves do not require a medium in which to travel. They can travel through a vacuum: the light from the Sun does this. They can also travel through matter. Light through air and glass, and x-rays through your body are other common examples.

8.5 Electromagnetic Waves

Electromagnetic waves travel at an extremely high speed. Their speed in a vacuum, called the “speed of light” because it was first measured using light, is represented by the letter c . Its value is:

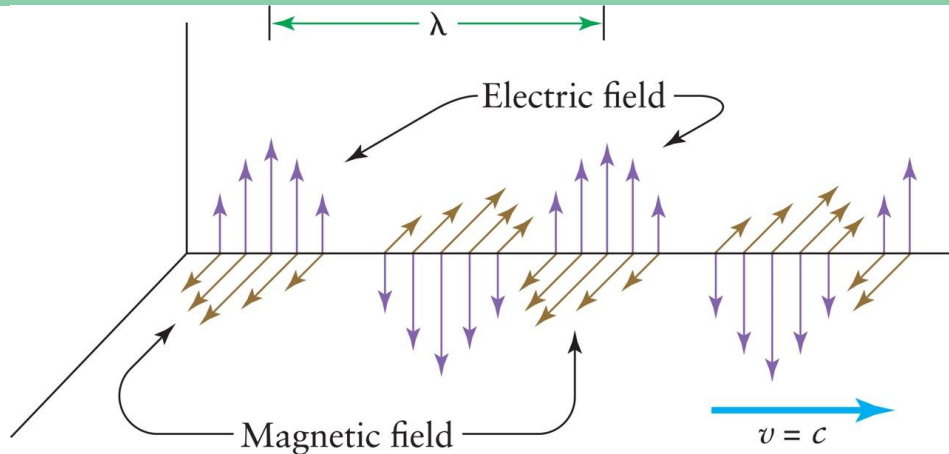
$$c = 299,792,458 \text{ m/s} \quad (\text{speed of light})$$

$$c = 3 \times 10^8 \text{ m/s} \quad (\text{approximately})$$

$$= 300,000,000 \text{ m/s}$$

$$= 186,000 \text{ miles/s} \quad (\text{approximately})$$

8.5 Electromagnetic Waves



All of the wave's variables we are already familiar with are applied to EM waves. The wavelength can be easily identified in the figure.

The equation $v = f \lambda$ holds with v replaced by c

$$c = f \lambda$$

Example 8.2

An FM radio station broadcasts an EM wave with a frequency of 100 megahertz. What is the wavelength of the wave?

The prefix mega stands for 1 million.

$$c = f \lambda$$

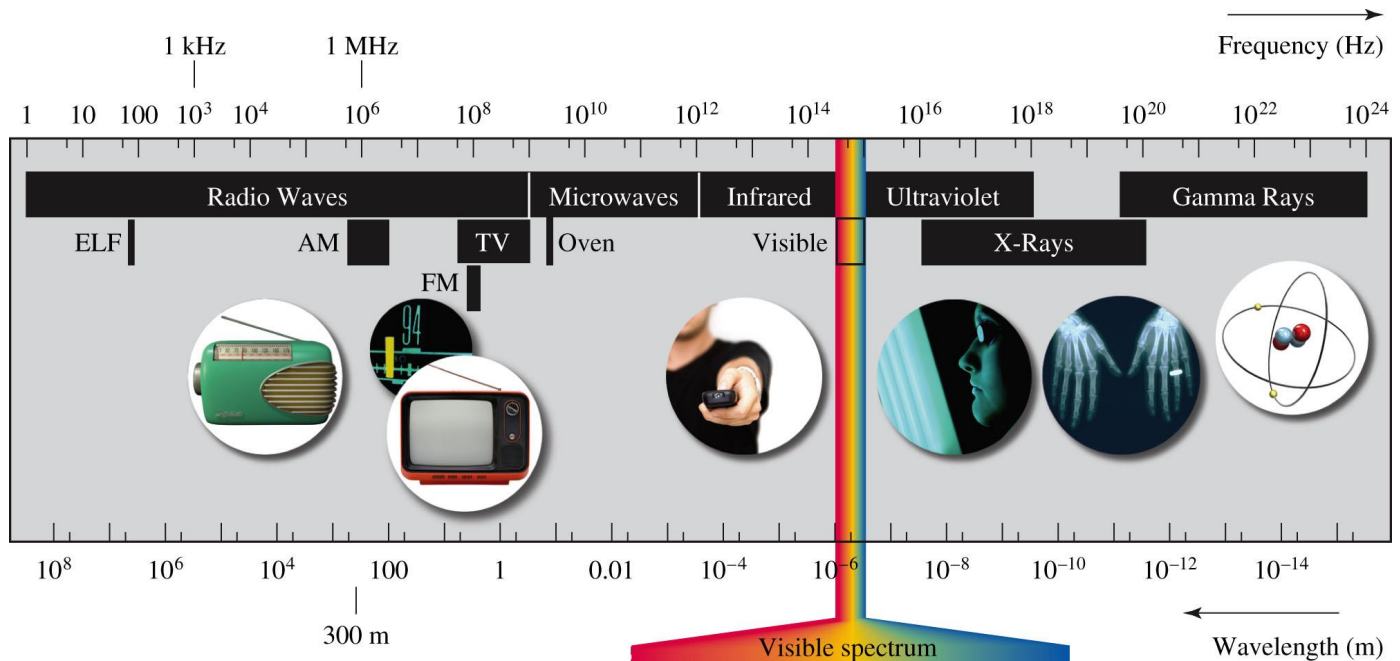
$$300,000,000 \text{ m/s} = 100,000,000 \text{ Hz} \times \lambda$$

$$\frac{300,000,000 \text{ m/s}}{100,000,000 \text{ Hz}} = \lambda$$

$$\lambda = 3 \text{ m}$$

8.5 Electromagnetic Waves

- *Electromagnetic waves* are named and classified according to frequency. In order of increasing frequency, the groups, or “bands,” are radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, x-rays, and gamma rays.
- Electromagnetic waves are a form of energy. Energy is needed to produce EM waves, and energy is gained by anything that absorbs EM waves.



8.5 EM: Radio Waves

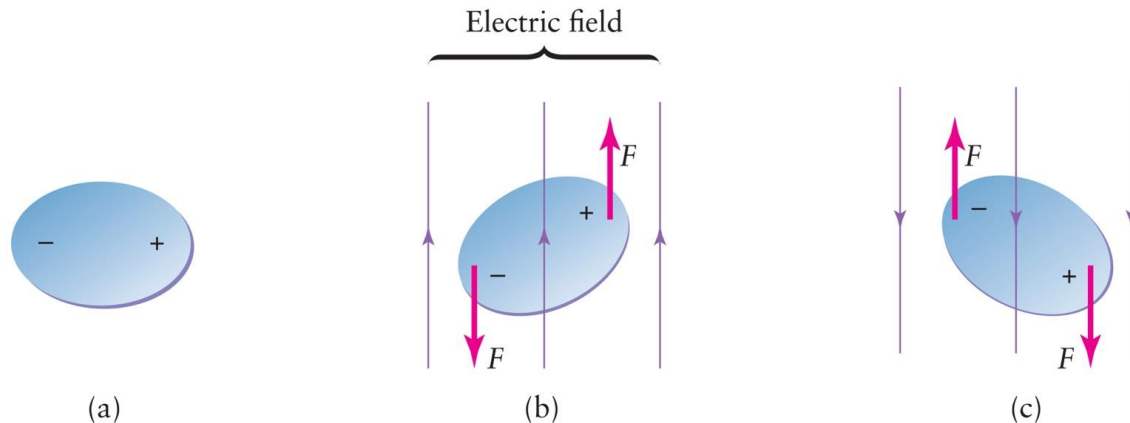
- *Radio waves*, the lowest frequency EM waves, extend from less than 100 Hz to about 1,000 megahertz.
- Within this range are a number of frequency bands that have been given separate names—for example, ELF (extremely low frequency), VHF (very high frequency), and UHF (ultrahigh frequency).
- By far the main application of radio waves is in communication. The process involves broadcasting a certain frequency of radio wave with sound, video, or other information “encoded” in the wave.



8.5 EM: Microwaves

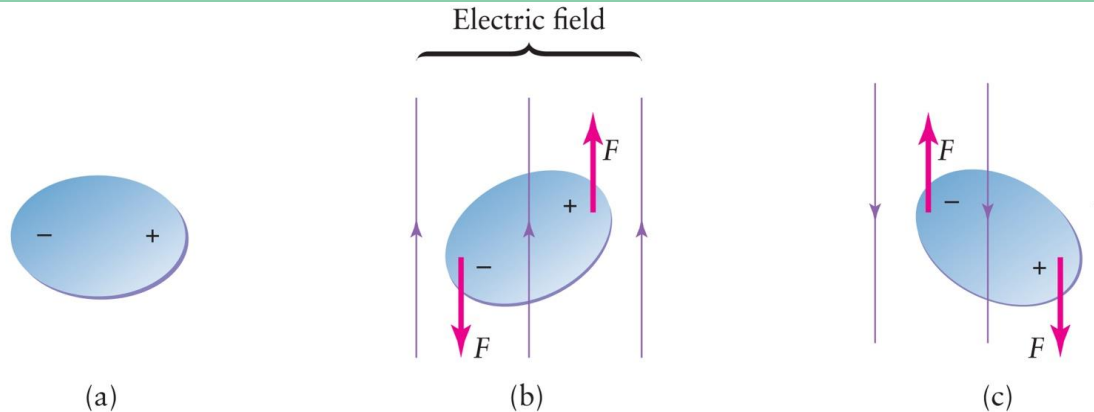
- The next band of EM waves, with frequencies higher than those of radio waves, is the *microwave* band.
- The frequencies extend from about 10^9 to 10^{12} hertz.
- One use of microwaves is in communication:
 - Bluetooth and WiFi signals that interconnect computers, cell phones.
 - Radar: using microwaves for echolocation.
 - Cooking food.

8.5 EM: Microwaves



- Microwaves have gained wide acceptance as a way to cook food. The goal of cooking is to heat the food—in other words, to increase the energies of the molecules in the food.
- Microwave ovens send microwaves into the food. The microwaves penetrate the food and raise the energies of the molecules directly.
- Recall that water consists of polar molecules—they have a net positive charge on one side and a net negative charge on the other side.

8.5 EM: Microwaves



- The electric field of a microwave exerts forces on the two sides of the water molecules in food. These forces are in opposite directions and twist the molecule.
- Because the electric field is oscillating, the molecules are alternately twisted one way and then the other. This process increases the kinetic energy of the molecules and thereby raises the temperature of the food.
- Cooking with microwaves is fast because energy is given directly to all the molecules. It does not rely completely on the conduction of heat from the outside to the inside of the food—a much slower process.

8.5 EM: Infrared

- ❑ **Infrared radiation** (IR; also called infrared light) has frequencies from about 10^{12} hertz to about 4×10^{14} hertz.
- ❑ Infrared radiation is ordinarily the main component of heat radiation.
 - Everything around you is both absorbing and emitting IR, just as you are.
 - The warmth you feel from a fire or heat lamp is the result of your skin absorbing the IR.
 - Infrared radiation is constantly emitted by atoms and molecules because of their thermal vibration. Absorption of IR by a cooler substance increases the vibration of the atoms and molecules, thus raising the temperature.
- ❑ IR is commonly used in wireless remote-control units for televisions and for short-distance wireless data transfer between such devices.



8.5 EM: Visible Light

- ❑ *Visible light* is a very narrow band of frequencies of EM waves that happens to be detectable by human beings. Certain specialized cells in the eye, called rods and cones, are sensitive to EM waves in this band.
- ❑ Within the narrow band of visible light, the different frequencies are perceived by people as different colors. The lowest frequencies of visible light are perceived as the color red. The highest frequencies are perceived as violet.

Color	Frequency Range ($\times 10^{14}$ Hz)	Wavelength Range ($\times 10^{-7}$ Hz)
Red	4.0–4.8	7.5–6.3
Orange	4.8–5.1	6.3–5.9
Yellow	5.1–5.4	5.9–5.6
Green	5.4–6.1	5.6–4.9
Blue	6.1–6.7	4.9–4.5
Violet	6.7–7.5	4.5–4.0

8.5 EM: Ultraviolet Radiation

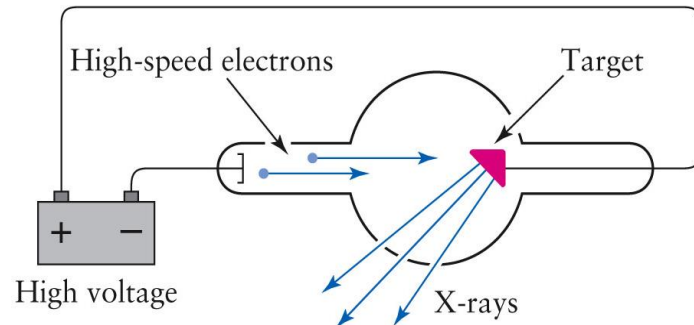
- ❑ *Ultraviolet (UV) radiation* has a frequency range from about 7.5×10^{14} hertz to 10^{18} hertz.
- ❑ Ultraviolet light is also part of the heat radiation emitted by very hot objects.
- ❑ About 7 percent of the radiation from the Sun is UV. This part of sunlight is responsible for suntans and sunburns.
- ❑ Ultraviolet radiation does not warm the skin as much as IR, but it does trigger a chemical process in the skin that results in tanning.

8.5 EM: Ultraviolet Radiation

- ❑ Overexposure to UV leads to sunburn as a short-term effect, and repeated overexposure during a person's lifetime increases the chance of developing skin cancer.
- ❑ UV radiation has many practical applications.
 - For example, it is used as an investigative tool at crime scenes to help identify bodily fluids such as blood.
 - Ultraviolet lamps are used to sterilize workspaces and tools used in biology laboratories and medical facilities.

8.5 EM: X-Rays

- ❑ The next higher frequency electromagnetic waves are *X-rays*. They extend from about 10^{16} to 10^{20} hertz.
- ❑ An important feature of x-rays is that their range of wavelengths (about 10^{-8} to 10^{-11} m) includes the size of the spacing between atoms in solids, therefore, it can be used to determine the arrangement of the atoms.
- ❑ X-rays also travel much greater distances through most types of matter compared to UV, visible light, and other lower-frequency EM waves.
- ❑ X-rays are produced by smashing high-speed electrons into a “target” made of metal. The electrons spontaneously emit x-rays as they are rapidly decelerated on entering the metal.



8.5 EM: X-Rays

- ❑ As x-rays pass through the body, Tissue containing elements with relatively large atomic numbers (Z), such as calcium ($Z = 20$), tend to absorb x-rays more effectively than those that contain predominantly light elements such as carbon ($Z = 6$) or hydrogen ($Z = 1$).
- ❑ Bones, which are rich in calcium, absorb x-rays better than soft tissue such as muscle or fat, and hence they show up more clearly on x-rays.
- ❑ X-rays (and gamma rays) can be harmful because they are ionizing radiation—radiation that produces ions as it passes through matter.
- ❑ Lead, with atomic number 82, is a particularly good shield for blocking x radiation.



8.5 EM: Gamma Rays

- ❑ The highest-frequency EM waves are *gamma rays*.
- ❑ The frequency range is from 3×10^{19} Hz to beyond 10^{23} Hz.
- ❑ The wavelength of higher-frequency gamma rays is about the same distance as the diameter of individual nuclei.
- ❑ Gamma rays are emitted in a number of nuclear processes: radioactive decay, nuclear fission, and nuclear fusion.

8.6 blackbody radiation

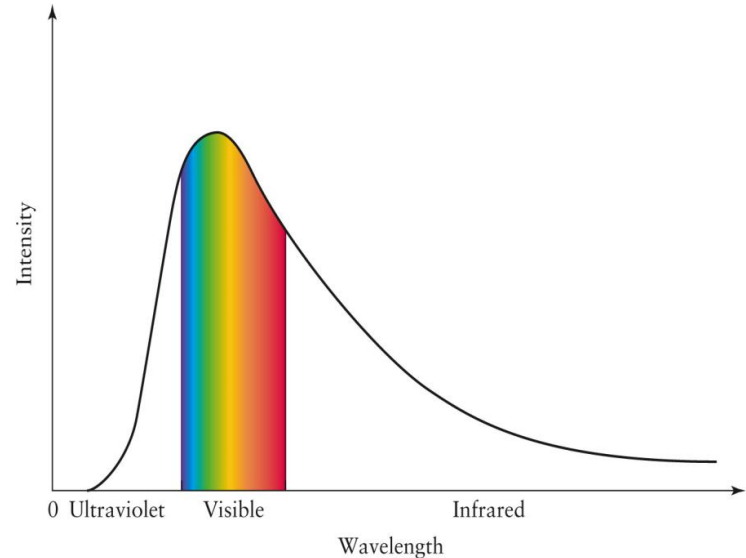
- Every object emits electromagnetic radiation because of the thermal motion of its atoms and molecules.
- The nature of the radiation emitted by a given object—the range of frequencies or wavelengths of EM waves present and their intensities—depends on the temperature of the object and on the characteristics of its surface (for example, its color).

8.6 blackbody radiation

- A hypothetical object that is *perfectly black—one that absorbs all EM waves that strike it*—would be the best at emitting heat radiation.
- Referred to as a **blackbody**, *it would emit radiant energy at a higher rate than any other object at the same temperature, and the intensities of all the wavelengths of EM waves emitted.*
- The heat radiation emitted by such an object is referred to as **blackbody radiation, or BBR.**
- Blackbody radiation, an idealized representation of heat radiation

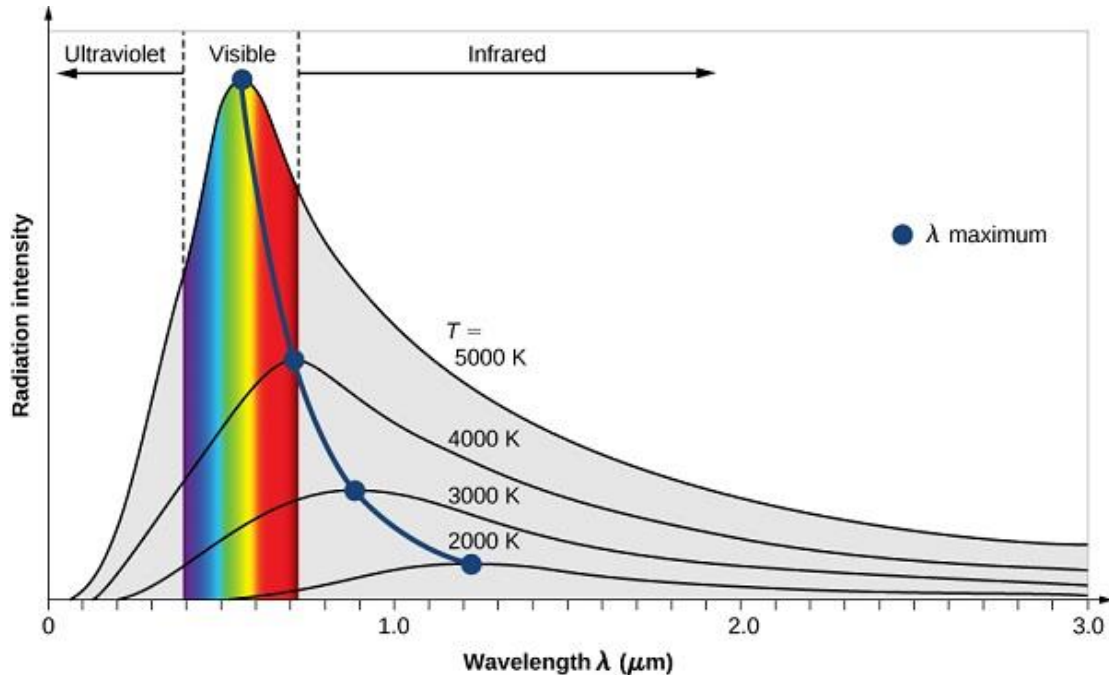
8.6 blackbody radiation

❑ The heat radiation emitted by any object (such as your own body, the Sun, or a blackbody) is a broad band of EM. Within this band, some wavelengths are emitted more strongly than others: the intensity of the different wavelengths, the amount of energy released per square meter of emitting surface per second, varies with wavelength.



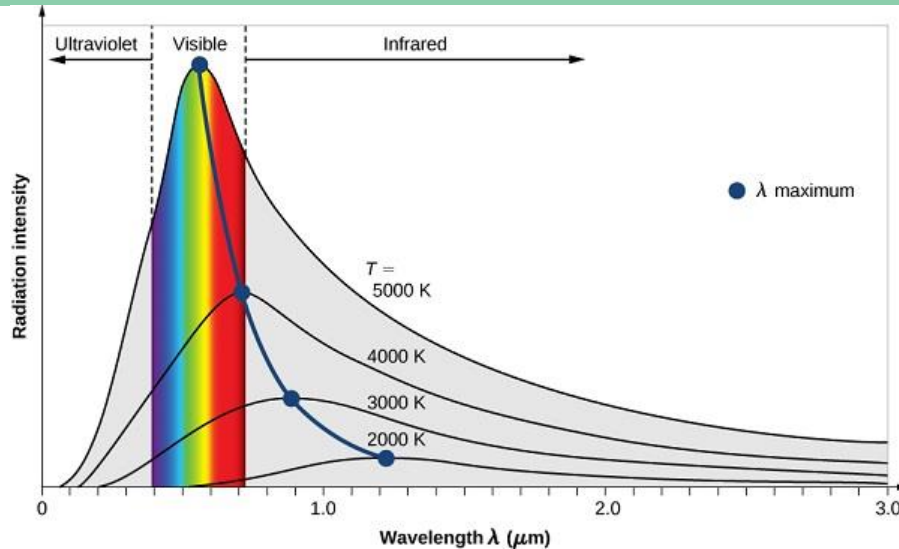
❑ The intensity of the visible wavelengths is higher than that of the IR wavelengths.

8.6 blackbody radiation



- ❑ A graph showing the intensity of each wavelength of radiation emitted by a blackbody is called a **blackbody radiation curve**.
- ❑ The size and shape of the graph change with the object's temperature.

8.6 Blackbody Radiation



Three aspects of heat radiation are affected by the object's temperature:

1. The total amount of each type of radiation (such as microwave and IR) emitted per second increases with temperature.

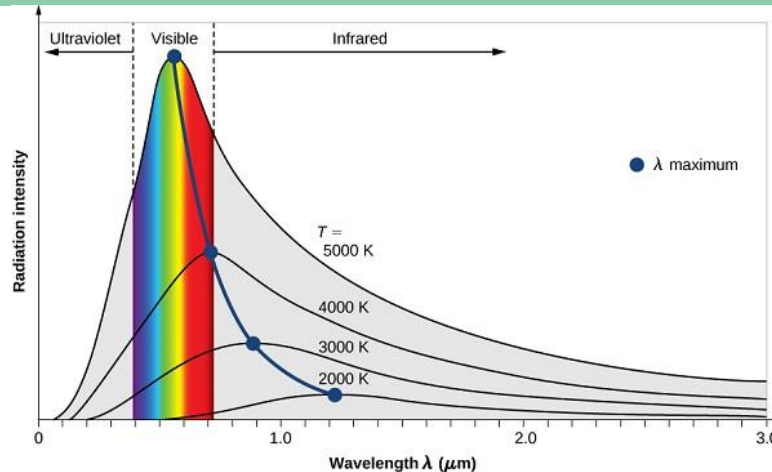
8.6 Blackbody Radiation

2. The total amount of radiant energy emitted per unit area per unit time increases rapidly with any increase in temperature. The total radiant energy emitted per second (power) is proportional to the temperature (in kelvins) raised to the fourth power.

$$P \propto T^4$$

- Doubling the Kelvin temperature of an object will cause it to emit 16 times as much radiant energy each second.

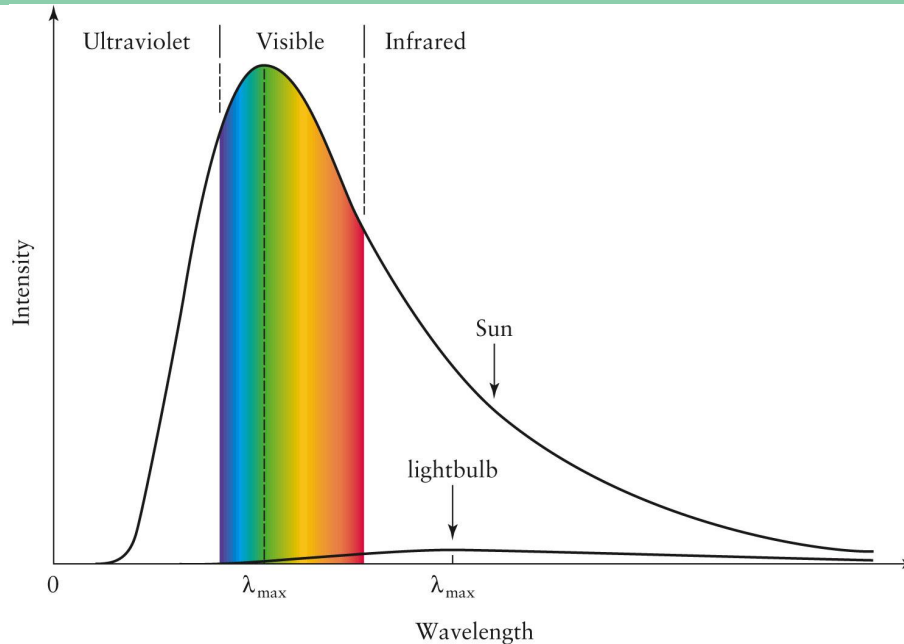
8.6 Blackbody Radiation



3. At higher temperatures, more of the power is emitted at successively shorter wavelengths (higher frequencies) of electromagnetic radiation.
 - For a blackbody, the wavelength that is given the maximum power (the peak of the blackbody radiation curve) is inversely proportional to its temperature.

$$\lambda_{\text{max}} = \frac{0.0029}{T} \quad (\lambda_{\text{max}} \text{ in meters, } T \text{ in kelvins})$$

8.6 Blackbody Radiation



- The filament of a lightbulb can be as hot as 3,000 K. The peak wavelength of its heat radiation curve is in the infrared region.
- At 6,000 K, the Sun's surface emits heat radiation that peaks in the visible band. It appears to be white hot.
- When comparing the curves, keep in mind that the Sun's is higher not because the Sun is bigger than a lightbulb but because it is hotter.

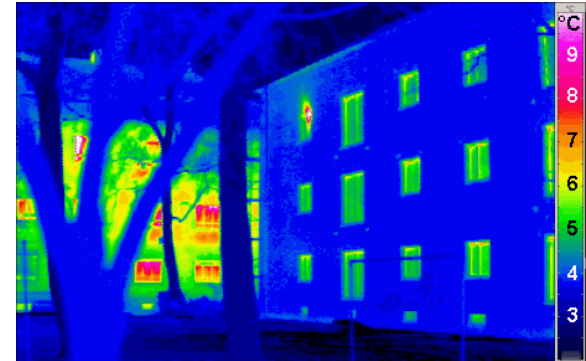
Example 8.3

Assuming that the Sun is a blackbody with a temperature of 6,000 K, at what wavelength does it radiate the most energy?

$$\lambda_{\text{max}} = \frac{0.0029}{T} = \frac{0.0029}{6,000 \text{ K}} = 4.8 \times 10^{-7} \text{ m}$$

8.6 Blackbody Radiation

- Most things on Earth have temperatures that cause them mainly to emit infrared light.
- Snakes use IR to hunt mice and other warm-blooded animals at night. These snakes have sensitive organs that detect the higher-intensity infrared emitted by objects warmer than their surroundings.
- Infrared-sensitive photographic film, video cameras, and other detection devices have many practical uses. For instance, IR photographs, called **thermograms**, can show where heat is escaping from a poorly insulated house. They can locate warmer areas on the human body that might be caused by tumors.
- The military uses IR detectors to locate soldiers at night, and doctors use them to determine patients' temperatures.

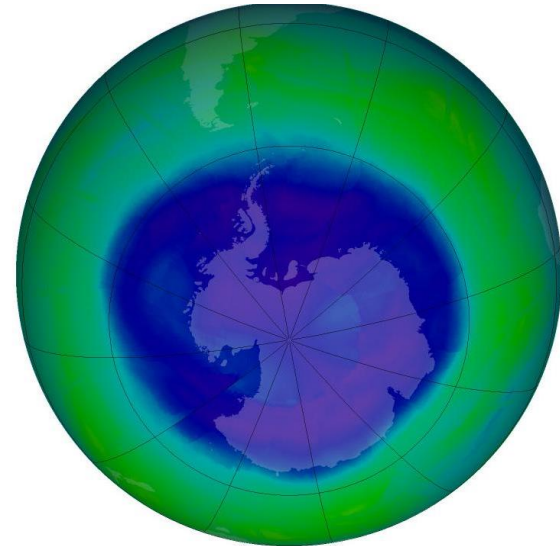


8.7 EM & Earth's atmosphere

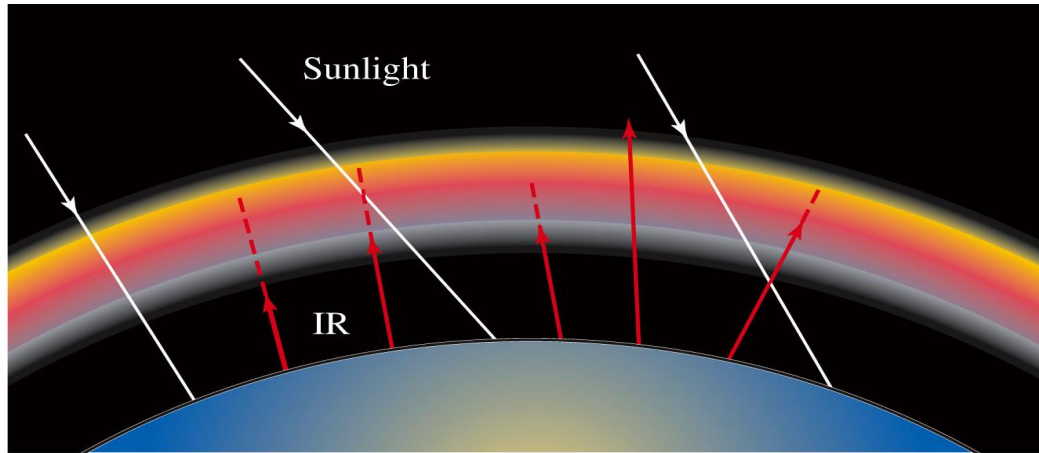
- ❑ Many substances in the atmosphere surrounding Earth interact with EM waves in important ways.
- ❑ In the region between about 20 and 40 kilometers above Earth known as the *ozone layer*, there is a comparatively high concentration of ozone, the form of oxygen with three atoms in each molecule.
- ❑ This ozone absorbs most of the harmful UV in sunlight. The ozone layer has been a shield protecting living things on Earth.

8.7 Earth's atmosphere: Ozone Layer

- It was reported that chlorofluorocarbons (CFCs), chemical compounds such as Freon used in refrigerators, air conditioners, and as an aerosol propellant in spray cans, could be depleting the ozone layer.
- The CFCs that are released into the air drift upward to the ozone layer and chemically break up the ozone molecules with alarming efficiency.
- Continued reduction in ozone levels could have tragic consequences:
 - Rates of occurrence of skin cancer could rise.
 - Crop yields could decrease, because increased UV adversely affects many plant species.

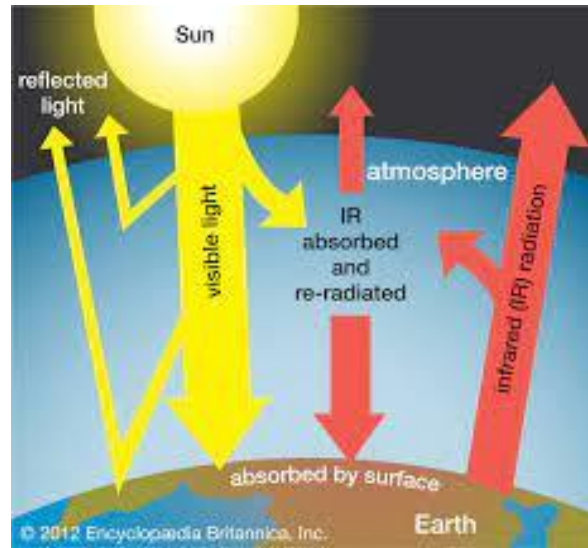


8.7 Earth's atmosphere: Greenhouse Effect



- The *greenhouse effect* is so named because it is partly responsible for keeping greenhouses warm in cold weather.
- A glass wall allows visible light to enter and to warm the interior.
- As the temperature of things inside increases, they emit more IR.
- Without the glass, this IR would escape from the interior and carry away the added energy. But the glass blocks the IR, so the added internal energy is trapped and the enclosure is warmed.

8.7 Earth's atmosphere: Greenhouse Effect

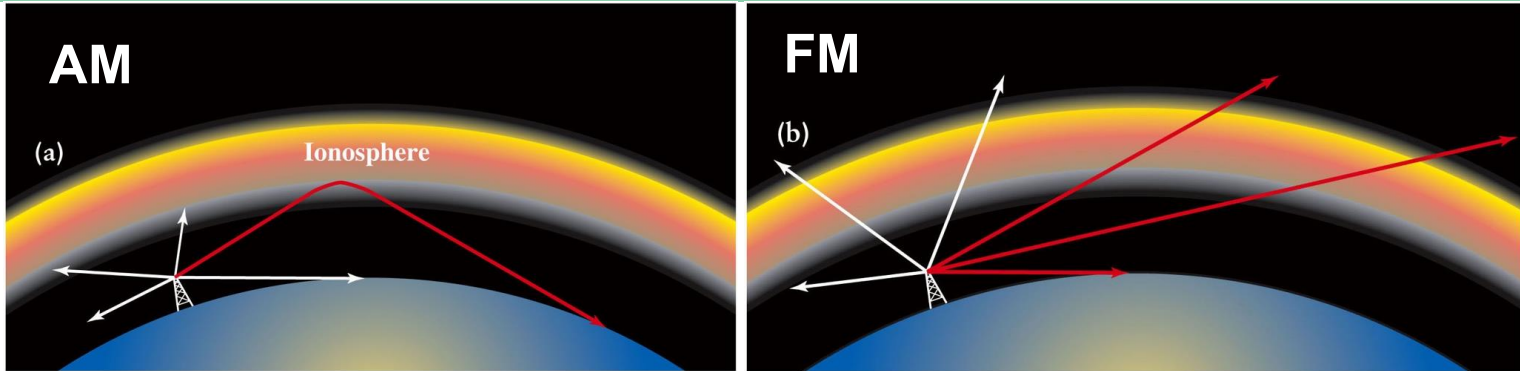


- The greenhouse effect occurs naturally in Earth's atmosphere.
- Water vapor, carbon dioxide, and other gases in the air act somewhat like glass in that they allow visible light from the Sun to pass through to Earth's surface while they absorb part of the IR emitted by the warmed surface and trap heat.
- The atmosphere is heated by the IR that is absorbed; it is about 35°C warmer than it would be without this effect.

8.7 Earth's atmosphere: Greenhouse Effect

- The atmospheric greenhouse effect is such a complex phenomenon that it is difficult to predict accurately what will happen. Different scenarios have been proposed:
 - As Earth heats up, huge amounts of carbon dioxide dissolved in the oceans could be released and could exacerbate the global warming.
 - Increased evaporation of water would raise the water vapor content of the atmosphere and possibly cause more heating.

8.7 Earth's atmosphere: The Ionosphere



About 50–90 kilometers above Earth's surface, in a region of the atmosphere known as the *ionosphere*, there is a relatively high density of ions and free electrons.

- Lower-frequency radio waves, used in AM radio, reflect off the ionosphere and return to Earth
- Higher-frequency radio waves, used in FM, pass through the ionosphere and out into space.

8.7 Earth's atmosphere: The Ionosphere



- ❑ The ionosphere is also the home of the auroras—the northern lights and the southern lights.
- ❑ Charged particles from the Sun excite atoms and molecules in the ionosphere, causing them to emit light.

Important Equations

IMPORTANT EQUATIONS

Equation	Comments
$\frac{V_o}{V_i} = \frac{N_o}{N_i}$	Input and output voltages of a transformer
$c = f\lambda$	Relates speed, wavelength, and frequency of EM waves
$\lambda_{\max} = \frac{0.0029}{T}$	Peak wavelength of BBR curve (SI units)
$P \propto T^4$	Power radiated by a blackbody with emitting area A (SI units)